

Brazil’s soy land footprint is shifting toward China and toward the agricultural frontier: appendices

Liza Buriak^{1*}

^{1*}Institute for Ecological Economics, Vienna University of Economics
and Business (WU), Vienna, Austria.

Corresponding author(s). E-mail(s): TODO@wu.ac.at;

Abstract

TODO: Appendices to the Data and Methods working document. Placeholder so
the file compiles standalone.

A Municipal soy accounting

Throughout the appendices, m and n index Brazil’s 5,570 municipalities, $p \in \{\text{bean, oil, cake}\}$ the three soy products, and i, j countries; matrices and vectors are set in bold. One operation recurs throughout the subnational reconstruction: a national total is distributed across municipalities in proportion to an observable weight, as listed in Table 2 of the Methods document. The released pipeline checks each such allocation against its national total at run time and halts on any violation.

A.1 Data assembly and domestic-use estimation

Five rules complete the allocation weights of Table 2 and the assembly of the municipal accounts:

1. exports declared without a municipality of origin are redistributed across the declared exporters of the same product and destination in proportion to their volumes, thus conserving national totals (undeclared imports are negligible);
2. the crushing-capacity weight is constructed by dividing each state’s reported capacity total equally among its active plants, as per-plant capacities are not published; the cost of this assumption is quantified in the Technical validation;
3. the biodiesel-capacity weight attributes each plant’s capacity to soy in proportion to its macro-region’s soy feedstock share;
4. municipal oil and cake output follow from the allocated crush: each year’s oil and cake yields are the ratios of the national oil and cake production to the crushed bean quantity in the FAO balance of that year, giving approximately 20% oil and 75% cake with a 5% processing loss, in line with the yields reported in the literature [1, 2];
5. all geometries are projected to SIRGAS 2000 / Brazil Polyconic (EPSG:5880); each municipality is represented by its administrative centre, and the straight-line distances between these centres provide the transport costs of Appendix D.

The treatment of stock changes, summarized in the last row of Table 2, is derived in Appendix C.

B Livestock systems and feed demand

B.1 Production-system disaggregation

IBGE municipal herd counts are divided into extensive and intensive chickens; extensive, intensive and industrial pigs; and grassland- and mixed-based cattle and buffalo, each split into dairy and meat. The municipal share of each system is the area-weighted zonal sum of the 2010 FAO gridded livestock rasters (GLW3) and the GLEAM ruminant production-system map, normalized across the systems of each species; where a municipality reports a herd in the run year but had no gridded animals in 2010, the state-mean share is substituted. Pigs follow the raster shares directly. For poultry, the extensive share of the total flock forms backyard birds, and the intensive share splits into layers (the reported laying hens) and broilers (the rest; all birds where layers are unreported). Feedlot cattle, reported only in the agricultural census, are the 2006 census counts scaled by the national confinement growth factor of the run year (e.g. $4.38/3.46 \approx 1.27$ for 2013, from ABIEC/Athenagro), holding the municipal pattern fixed; the remaining cattle are split into grassland and mixed systems by the raster shares, with dairy identified by milked cows and meat as the residual. Buffalo use the municipal milked-cow ratio of cattle as the dairy proxy. Each species' subsystems partition its reported herd exactly.

B.2 Feed demand and national rescaling

Each herd is multiplied by its GLEAM per-animal intake (annual dry-matter intake times the share sourced from soybeans and soy cake, converted to as-fed tonnes at 88% dry matter). One national factor per product then rescales the bottom-up totals to the FAO national feed figures, preserving the spatial and cross-system distribution. Municipal feed demand for beans and cake is the sum over systems. Holding the 2010 system shares fixed over the study period is a simplification: herd sizes remain annual, misclassification affects only how a herd is split across intake intensities, and the national rescaling bounds the aggregate error, which grows with distance from the 2010 reference year.

C National balancing and trade reconciliation

For each product, the national commodity balance imposes the supply–use identity: production, imports and stock withdrawals (supply) equal exports, food, feed, seed, processing and other uses (use). The trade harmonisation collapses destinations absent from the FABIO/FAOSTAT bilateral list into a rest-of-world aggregate; it aligns classifications only and applies no rescaling. The balance items are rescaled proportionally, with each item's municipal distribution as weight, after the national residual has been moved into the stock term. A net national stock withdrawal is carried on the supply side: the drawdown is distributed across municipalities by the stock weights (storage capacity for beans, production for oil and cake) and added to municipal supply. A drawdown is a genuine source of current-year supply, and this treatment alone keeps every municipal use non-negative, as required by the re-export inversion of Appendix E. Net additions remain on the use side. As each year is solved independently, tonnes supplied from a drawdown, physically grown in earlier years, inherit the current year's soy-area intensity (Appendix G). Finally, each municipality's product balance defines its surplus, $\max(\text{supply}_m^p - \text{use}_m^p, 0)$, and deficit, $\max(\text{use}_m^p - \text{supply}_m^p, 0)$, which drive the inter-municipal flows of Appendix D; the two totals match by construction after the rescaling, as required by the balanced transportation problem. The national residual itself cannot be resolved into flows; it is retained as FABIO's signed balancing column and re-enters the model as a dedicated final-demand category (Appendices F and G).

D Optimal-transport formulation

D.1 The Euclidean model

For each product p , the inter-municipal flow matrix $\mathbf{F}^p$ solves the balanced transportation problem

$$\mathbf{F}^p = \arg \min_{\mathbf{F} \geq 0} \sum_{m,n} F_{mn} d_{mn} \quad \text{s.t.} \quad \sum_n F_{mn} = \text{surplus}_m^p, \quad \sum_m F_{mn} = \text{deficit}_n^p, \quad (\text{D.1})$$

where d_{mn} is the straight-line distance between the administrative centres of municipalities m and n (Appendix A) (the two margins are balanced beforehand by proportionally scaling the larger one). The problem is solved exactly by the network-simplex algorithm (R package `transport` [3]), separately for beans, oil and cake.

D.2 The multimodal network variant

To test whether the straight-line metric distorts the allocation, the same municipal surpluses and deficits are re-solved on a multimodal representation of Brazil’s logistics network, generalizing the single-year (2013) intermodal model of Trsek [4] to the years 2010–2022. The variant has three building blocks:

1. the network: roads (OpenStreetMap, simplified to a 5 km grid), railways (ANTT lines and stations) and waterways (DNIT rivers with ANTAQ ports). A tonne may travel by road directly, or by road to a rail or waterway terminal, along that network, and onwards by road to its destination;
2. the costs: every leg is priced at a unit cost per tonne-km that depends on the mode (road 0.013–0.174, rail 0.006–0.065, water 0.004–0.032 USD), and changing modes carries a switching charge (0.74–2.57 USD per tonne); the central runs use the midpoints of these ranges, taken from the freight-cost literature compiled in [4];
3. the capacities: a rail or waterway corridor cannot carry more of a product than it was observed to carry, from ANTT origin–destination rail records (all study years) and ANTAQ port statistics (observed from 2017 onwards; earlier years are proxied with the 2017 records, as ANTAQ’s pre-2017 archives are currently unavailable).

The resulting minimum-cost flow problem is solved with the open-source HiGHS solver; the 2021 problem is infeasible on the current inputs and is excluded from the comparison.

E Origin tracing and re-export disentangling

E.1 Linking origin municipalities to importing countries

The transport flows of Appendix D record only what moves between municipalities, so the part of each municipality’s supply that stays and is used at home is added on the diagonal,

$$\tilde{F}_{mn}^p = F_{mn}^p + \mathbf{1}[m = n] (\text{supply}_m^p - \text{surplus}_m^p), \quad (\text{E.1})$$

so that the rows of $\tilde{\mathbf{F}}^p$ sum to total municipal supply and its columns to total municipal use. Every exported tonne is then unwound to its origin (Figure 3),

$$X_{mj}^p = \delta_m^p \sum_n \phi_{mn}^p E_{nj}^p, \quad (\text{E.2})$$

where E_{nj}^p are the observed exports of municipality n to country j ; the sourcing share $\phi_{mn}^p = \tilde{F}_{mn}^p / \sum_{m'} \tilde{F}_{m'n}^p$ is the fraction of n ’s total use that originated in municipality m (including $m = n$); and the domestic-origin share $\delta_m^p = (\text{supply}_m^p - \text{imports}_m^p) / \text{supply}_m^p$ strips out quantities of foreign origin (withdrawal-supplied tonnes count as domestic under the supply-side stock treatment; Appendix C). Local consumption is traced with

the same formula through a synthetic domestic column added to the export table (for beans net of the crush, which is traced through the oil and cake chains instead), and oil and cake are traced one step further back, to the bean-growing municipalities, by premultiplying their origin-to-destination matrices with the bean sourcing shares and the bean domestic-origin share.

E.2 Re-export disentangling

To attribute internationally traded soy to its true origin, bilateral trade is disentangled with the standard Leontief-type inversion of physical trade-linkage accounting [5]: trade is normalized by destination use, the infinite re-export chain is summed by matrix inversion, and retaining the domestically produced quantities yields the origin-to-consumption matrix,

$$\mathbf{R} = \mathbf{T} \hat{\mathbf{u}}^{-1}, \quad (\mathbf{I} - \mathbf{R})^{-1} = \sum_{k=0}^{\infty} \mathbf{R}^k, \quad (\text{E.3})$$

where $\mathbf{T}$ is the bilateral trade matrix, $\hat{\mathbf{u}}$ the diagonalized vector of total use by region, R_{ij} the share of region j 's use supplied by imports from region i , and $\mathbf{R}^k$ carries the flows re-exported k times, so the inverse sums chains of every length. The departure from the published method is the regional dimension: for the soy products Brazil is replaced by its 5,570 municipalities, enforced throughout the processing of the trade data so that no national Brazilian record re-enters the inversion. The inversion treats the soy available in a region as one well-mixed pool: local production and incoming shipments blend in proportion to their volumes, and every user of the pool receives the same origin mix, as is natural for a bulk commodity blended in silos. Which municipality exports to which country is observed in the customs data, so the assumption does not touch that step; it fills in only where an exporting municipality's soy came from within its catchment, and how soy is passed on when a country re-exports it. What it cannot represent is a buyer who deliberately selects soy of a specific origin, as European importers do under the Amazon Soy Moratorium: such origin-specific contracting is not resolved. The inversion is computed exactly by a sparse solver, with a dense fallback and, as a last resort, a small diagonal ridge to lift a genuine unit eigenvalue (a commodity whose recorded global trade is entirely re-exported); the years requiring the ridge are documented with the released code.

F FABIO backends and temporal coverage

The FABIO v2 build exists only from 2010 onwards; for 2001–2009 the pipeline switches to four inputs from the FABIO v1.1 build: the full bilateral trade detail, the land-use extension, the feedstock optimisation results, and the v1 balance sheets (the v2 balance file contains only placeholder rows before 2010). Version bridges map through stable FAO item codes, and the v1.1 land-use extension, which reports total agricultural land rather than v2's cropland, is converted to the per-process intensity vector through the same mapping; soy is cropland under both definitions, so the soy land footprint is unaffected, but non-soy land intensities are not comparable across vintages. The seam is quantified on the overlap years 2010–2013, run through both chains end to end: in the two clean overlap years totals differ by -6% (2011) and -8% (2013, v1.1 low) with cross-country correlations of consumer footprints of 0.988 and 0.992 (the 2012 v1.1 build halts at the re-export conservation check, and 2010 diverges through a disagreement between the balance-sheet vintages), so pre-2010 levels are reported unadjusted. One usage caveat: Brazil's domestic footprint is about a third lower on v1.1, so cross-vintage statements about Brazil-domestic use are restricted to 2010 onwards.

G Land-use footprint accounting

G.1 Footprint calculation

With $e_i = \text{land}_i/x_i$ the land intensity of process i (its direct soy area per unit of output; Section 2.6), the footprint is the standard intensity–Leontief–demand product

$$\mathbf{H} = \hat{\mathbf{e}} \mathbf{L} \mathbf{Y}, \quad (\text{G.1})$$

where $\hat{\mathbf{e}}$ is the diagonal matrix of the land intensities, $\mathbf{L}$ the Leontief inverse of the nested system, $\mathbf{Y}$ final demand aggregated to consumer countries, and H_{ij} the hectares of soy land in origin municipality i embodied in country j ’s consumption. Stock additions and the balancing category are removed from final demand before aggregation: stock drawn down in year t embodies land use of earlier years, and retaining it as negative year- t demand would attribute negative footprints to producer regions in drawdown years; EXIOBASE inventory changes are treated analogously. The food system and the non-food background traced through EXIOBASE [6] contribute additively to each consumer’s total. Footprints by consumed product are obtained by restricting $\mathbf{Y}$ to selected commodity groups (e.g. dairy, meat) before applying (G.1).

G.2 Origin attribution and grid refinement

The biome assignment intersects each municipality’s interior point with the biome layer; points falling outside every biome polygon are assigned the nearest biome. A destination’s frontier share is the fraction of its origin footprint located in frontier municipalities, with the frontier as defined in Section 2.6. For visualization, per-municipality embodied production can optionally be distributed onto 30 m MapBiomas soy pixels, each pixel of a municipality carrying that municipality’s per-pixel share for the given consumer country; this refinement is illustrative and does not affect the reported results.

H Validation and uncertainty analysis

H.1 Benchmark and baseline

The comparison sample is built from the origin-to-importer flows X_{mj}^p of Appendix E, the Trase composite, and the downscaling baseline, defined as national exports to each destination allocated to municipalities in proportion to their share of domestic-origin soybean supply: every importer draws from the same municipal map, importers differ only in total volume, and the baseline contains no supply-chain structure by construction. All three sources are converted to bean equivalents with the national crush ratio and summed over beans, oil and cake. Trase flows with unknown origin municipality are excluded, destinations are harmonized to the bilateral-trade classification, and the sample comprises all soy-producing municipalities crossed with the destinations recording positive exports in both benchmark and model. Brazil, which enters the tables as the destination of domestic use, is excluded wherever importers are compared or weighted.

H.2 Agreement by importer

Figures 1 and 2 unpack the volume-weighted average reported in the Technical validation section at the two ends of the volume distribution: the four importers that together take 70% of the Trase-recorded 2010–2022 volume (China, the Netherlands, Thailand and Spain), and the five smallest importers with matched flows in every year (Cuba, Israel, the United Arab Emirates, Greece and Australia, 0.1–1.5 Mt each over the whole period). Each figure shows the mean correlation over its importer set and, below it, the full per-importer scores per year for the three allocation variants.

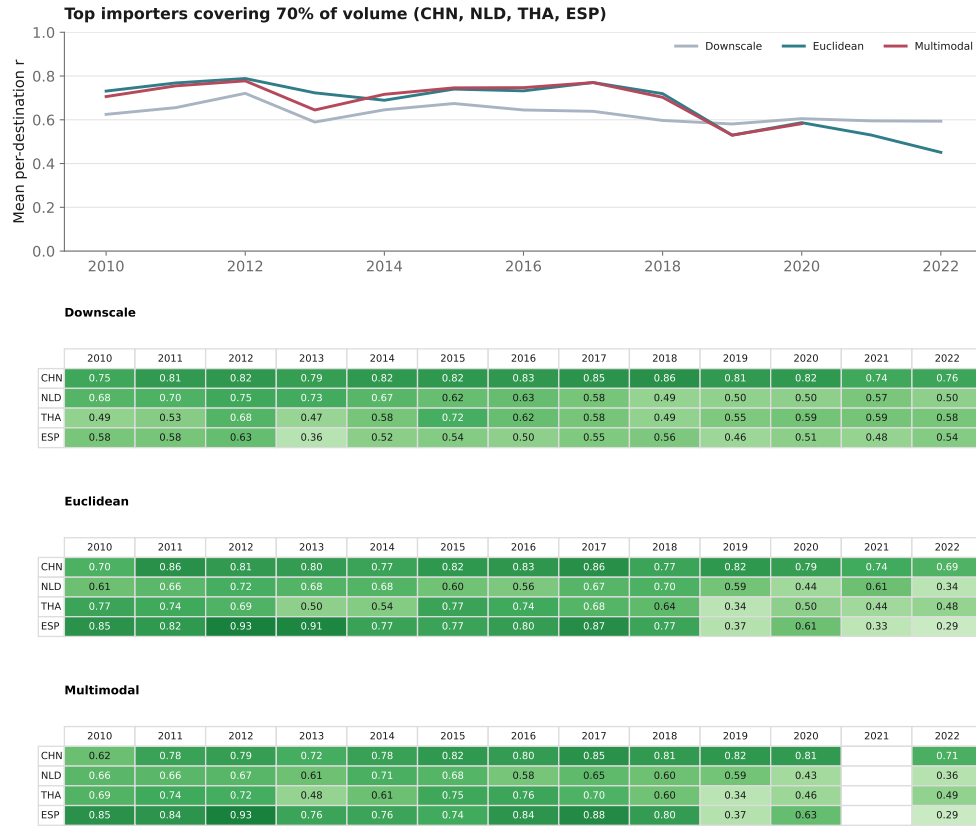


Fig. 1 Per-destination correlation with Trase for the four importers covering 70% of the 2010–2022 volume. The line plot shows the mean over the four importers; the tables give the per-importer scores per year for each allocation variant (colour scale from 0, white, to 1, dark green; the multimodal variant has no 2021 solve).



Fig. 2 As Figure 1, for the five smallest importers matched in every year. At this volume level, three orders of magnitude below China, the correlations scatter from slightly negative to above 0.9 with no persistence from year to year: individual small-importer results carry little information.

H.3 Perturbation recipes of the input-sensitivity screen

Crush location. Each state’s fixed capacity total is redistributed over its plant municipalities in proportion to a weight k_m : one in the baseline (equal split), municipal soybean production in the production-proportional variant, and $k_m \sim \text{Exp}(1)$ drawn independently per plant in the random variants, which makes every split of the state total equally likely (a uniform Dirichlet distribution over the shares); ten seeds give ten capacity maps. The allocation-sharpness variants leave the split untouched and raise the allocation weight to cap_m^γ with $\gamma \in \{0.5, 2\}$.

Feed rates. The per-head bean and cake rates of every livestock system are multiplied by independent mean-one lognormal factors,

$$f = \exp(\mathcal{N}(-\sigma^2/2, \sigma^2)), \quad \sigma = 0.3, \quad (\text{H.1})$$

drawn separately for the bean and the cake rate of each system; ten seeds give ten parametrisations. Because national feed totals are rescaled to the FAO commodity balance after the multiplication (Appendix B), a factor common to all systems cancels exactly, and only the relative system mix moves the spatial allocation. The herd-composition variant replaces the 2010-frozen municipal system shares with state averages.

H.4 Volume–correlation decision rule

The volume floor arises because small volumes are handled by few shipments and carry little spatial information, so per-destination agreement degrades for all methods alike; Figure 10 shows the flow-level pattern for four sample years. The resulting decision rule (model subnationally above the floor, downscale below it) is reported with the benchmark tables of every study year, and results for individual small-volume destinations should be read with caution.

I Data access links

Dataset names below match Table 1, which gives provider, role and coverage; this annex adds only the access links. All URLs were last verified in July 2026. Brazilian government portals may reject automated requests but resolve in a browser. A programmatic downloader covering the machine-retrievable sources ships with the code (`code/download_data.py`).

Brazilian subnational statistics

- **Soy production, area** (SIDRA table 1612): <https://sidra.ibge.gov.br/tabela/1612>; API pattern <https://apisidra.ibge.gov.br/values/t/1612/n6/all/v/109,216,214/p/{year}/c81/2713>.
- **Population** (tables 6579/9514), **herds** (table 3939), **milk cows** (table 94), **feedlot cattle** (tables 919/6911): same API pattern with the respective table number.
- **Grain storage** (table 278): <https://sidra.ibge.gov.br/tabela/278>.
- **Feedlot confinement series** (ABIEC Beef Report; manual download): <https://www.abiec.com.br/publicacoes/>.
- **Crush/refining capacity** (manual download): <https://abiove.org.br/estatisticas/>.
- **Biodiesel capacity** (Statistical Yearbook table 2.6; manual download): <https://www.gov.br/anp/pt-br/centrais-de-conteudo/publicacoes/anuario-estatistico/>.
- **Soy-oil acquisition (POF)**: <https://www.ibge.gov.br/estatisticas/sociais/educacao/9050-pesquisa-de-orcamentos-familiares.html>.
- **Municipal trade**: per-year files <https://balanca.economia.gov.br/balanca/bd/comexstat-bd/mun/> (`EXP_{year}_MUN.csv`, `IMP_{year}_MUN.csv`); lookup tables <https://balanca.economia.gov.br/balanca/bd/tabelas/> (`PAIS.csv`, `UF_MUN.csv`). Directory listings are blocked; the files themselves download normally.

National balances and feed parameters

- **Commodity balance (FBS)**: <https://www.fao.org/faostat/en/#data/FBS>; bulk file [https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_All_Data_\(Normalized\).zip](https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_All_Data_(Normalized).zip).
- **Detailed trade matrix**: bulk file [https://bulks-faostat.fao.org/production/Trade_DetailedTradeMatrix_E_All_Data_\(Normalized\).zip](https://bulks-faostat.fao.org/production/Trade_DetailedTradeMatrix_E_All_Data_(Normalized).zip).
- **Gridded livestock (GLW3)**: <https://dataverse.harvard.edu/dataverse/glw>.
- **Production systems, feed (GLEAM)**: <https://www.fao.org/gleam/resources/en/>.

Transport network layers

- **Road network**: <https://download.geofabrik.de/south-america/brazil.html>.
- **Waterways**: <https://www.gov.br/dnit/pt-br/assuntos/atlas-e-mapas>; download tool VGeo <https://servicos.dnit.gov.br/vgeo/> (may reject automated requests).
- **Rail lines, stations, cargo**: <https://dados.antt.gov.br/dataset/transporte-ferroviario-de-cargas-e-passageiros>.
- **Ports, port cargo**: <https://web.antaq.gov.br/anuario/> (temporarily unreachable at the July 2026 check; re-verify before use).

Geographies

- **Municipal boundaries** (geobr): <https://github.com/ipeaGIT/geobr>; <https://cran.r-project.org/package=geobr>; localities API <https://servicodados.ibge.gov.br/api/v1/localidades/municipios>.
- **State boundaries**: <https://gadm.org/>.
- **Biomes 1:250k**: <https://www.ibge.gov.br/geociencias/informacoes-ambientais/estudos-ambientais/15842-biomas.html>.
- **Soy land tiles** (via Google Earth Engine): <https://mapbiomas.org/>; export toolkit <https://github.com/mapbiomas-brazil/user-toolkit>.

Global models and reference series

- **FABIO v2 / v1.1**: <https://github.com/fineprint-global/fabio>; v1.1 archive DOI 10.5281/zenodo.2577066.
- **EXIOBASE 3 pxp**: <https://www.exiobase.eu/>; data DOI 10.5281/zenodo.3583070.
- **Trase composite v2.6.1**: <https://trase.earth/>; supply-chain dataset DOI 10.48650/DCE3-JJ97 (v2.6).

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- [6] Stadler, K. *et al.* EXIOBASE 3: Developing a time series of detailed environmentally extended multi-regional input-output tables. *Journal of Industrial Ecology* **22**, 502–515 (2018).